

Uncertainty intervals

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Sampling variability

Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Experimental units and independence

Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

A literature comparison matrix

Without opening the old lesson, explain one key idea from A literature comparison matrix. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Uncertainty intervals. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Uncertainty intervals. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Mean 20, $n=25$, known $\sigma=5$ and critical 1.96. Compute interval and name one assumption.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Observed mean 10, $n=16$, $\sigma=2$. Compute standard error, margin and interval. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A narrow interval can be precisely biased; sampling independence and representativeness must be justified. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Mean 20, $n=25$, known $\sigma=5$ and critical 1.96. Compute interval and name one assumption.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled uncertainty intervals evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of uncertainty intervals. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Uncertainty intervals

Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.

Estimated difference $B - A$ is -0.8 seconds; supplied 95% confidence interval is $[-1.4, -0.2]$ seconds. Assume the study's interval method is appropriate.

Supplied interval for B minus A time

	Lower endpoint (s)	Estimate (s)	Upper endpoint (s)
Original supplied interval	-1.4	-0.8	-0.2
Changed supplied interval	-1.4	not specified	0.3

Read a cell using both its row and column labels.

Uncertainty intervals

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Locate the estimate

-0.8 lies between the interval endpoints.



2. Interpret the sign

Negative B-A means B took less time under this measurement convention.



3. Respect coverage meaning

95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

CHANGE ONE THING

Use a supplied interval $[-1.4, 0.3]$ instead.

Explain the changed result using the matching lesson.

Locate -0.8 and zero relative to the supplied interval $[-1.4, -0.2]$. Interpret the sign and confidence-method meaning, then describe what changes with interval $[-1.4, 0.3]$.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

1. Interpret the sign
2. Locate the estimate
3. Respect coverage meaning

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Respect coverage meaning 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter. An uncertainty interval conveys precision under assumptions. Its endpoints, units, comparison direction and method all matter; the interval is supplied here, not calculated from unavailable raw data.

Practice 2: explain the deciding evidence

Changed-case practice: Use a supplied interval $[-1.4, 0.3]$ instead. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. It includes zero and positive differences, so both no difference and slower B remain compatible with that

interval.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval. The new upper endpoint crosses zero.

Repair a missing building block

Review Sampling variability (ai-7/11). Goal: Explain the mechanism and terminology of sampling variability. Short practice: Calculate the two four-case rates and their difference. Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.. Return to this week and explain what you changed.

Review Experimental units and independence (ai-7/9). Goal: Explain the mechanism and terminology of experimental units and independence. Short practice: Count independent plants and photo views. Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.. Return to this week and explain what you changed.

Review A literature comparison matrix (ai-7/6). Goal: Explain the mechanism and terminology of a literature comparison matrix. Short practice: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the literature matrix preserve?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
